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Final Year Project
Software Requirements Specification
For
Export Edge AI
Version 1.0

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1. Purpose

This Software Requirements Specification (SRS) document details the functional and non-functional requirements for the "Export Edge AI" system. The system is an integrated agri-tech solution designed to transform the mango exporting business by reducing post-harvest loss and enhancing market competitiveness using AI-powered quality control and real-time market intelligence. This SRS covers the full scope of the project from data collection to deployment.

1.2. Document Convention

This SRS follows standard conventions for technical documentation. Headings and subheadings are numbered hierarchically (e.g., 1, 1.1, 1.2), and functional requirements are clearly marked using bullet points or numbered lists for easy reference. Special terms and keywords are italicized for emphasis. The document assumes that higher-level requirements will inherit the priority of more detailed requirements, unless explicitly stated otherwise. In terms of formatting:

- Bold text represents section titles.
- Italics denote important terms or acronyms.
- Inline code and commands will be in monospace font.

1.3. Intended Audience and Reading Suggestions

This document is intended for the following stakeholders:

- **Developers:** To understand system functionalities, requirements, and design.
- **Project Managers:** To monitor progress, ensure deliverables, and manage timelines.
- **Testers:** To create test cases and validate system functionality against requirements.
- **End Users:** Urban planners and government agencies interested in utilizing the application.
- **Documentation Writers:** To prepare user manuals and technical documentation.

For a general understanding, all readers should start with the **Introduction**. Developers and testers should focus on **Functional Requirements** and **System Features**, while project managers may prioritize **Project Milestones and Deliverables**.

1.4. Product Scope

ExportEdge is a deep learning-based solution that integrates computer vision and large language models (LLMs) to accurately classify mangoes and recommend the best possible export price and destination country. By leveraging state-of-the-art techniques such as Convolutional Neural Networks (CNNs) and Retrieval-Augmented Generation (RAG), the system automatically generates comprehensive mango

export reports for producers and exporters. ExportEdge aims to minimize post-harvest losses and support global trade efficiency, directly contributing to the United Nations Sustainable Development Goals (SDGs) — particularly **Goal 8 (Decent Work and Economic Growth)**, **Goal 9 (Industry, Innovation and Infrastructure)** and **Goal 12 (Responsible Consumption and Production)**.

1.5. References

1. United Nations. *Sustainable Development Goals (SDG 2: Zero Hunger & SDG 8: Decent Work and Economic Growth)*. United Nations, 2023. Available at: <https://sdgs.un.org/goals>
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2. Overall Description

2.1. Product Perspective

ExportEdge is a new, self-contained product developed to assist mango exporters and producers in maximizing profits and reducing post-harvest losses. The product leverages mango images, deep learning models, and Retrieval-Augmented Generation (RAG) to generate an interactive, real-time mango analysis report. **ExportEdge** does not aim to replace any current solution but instead fills a critical market gap by offering an affordable and scalable approach for mango export optimization. The product will be developed as a **web-based platform** that connects an **LLM module** and a **camera interface** to analyze mangoes in real time. It will integrate the following major components:

- **Deep Learning Models:** Classification and segmentation models for detecting mango quality and diseased regions.
- **Mango Image Interface:** A camera-based module for capturing real-time images of mangoes on the conveyor system.
- **User Interaction Layer:** A web interface for uploading mango images, viewing classifications, and generating export reports through the LLM.
- **Data Pipeline:** An automated process for image preprocessing, inference, and report generation.

The following context diagram (Figure 1) illustrates the overall structure and interaction among these system components.

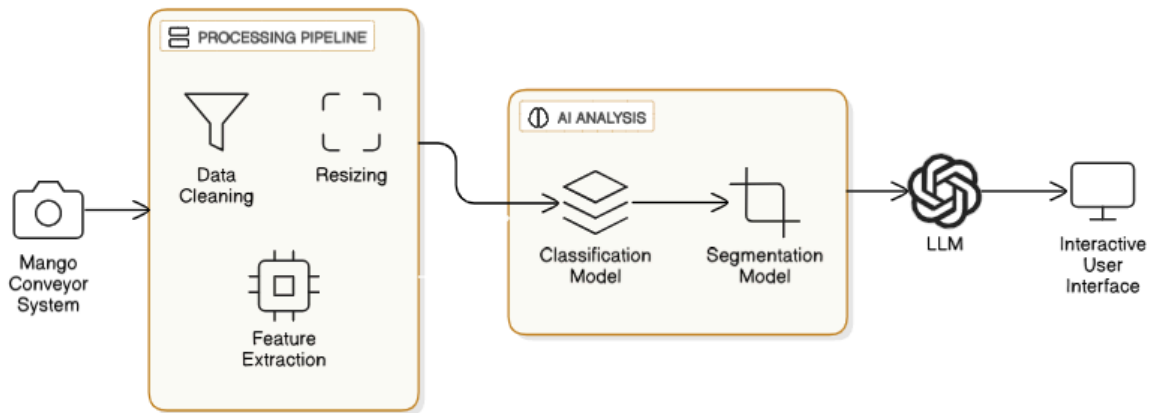


Figure 1. System context diagram of Export Edge showing interaction between different modules and end users of the system

2.2. Product Functions

The product will provide the following key functionalities:

- **Mango Image Collection:** A camera integrated with the conveyor belt will automatically capture high-quality images of mangoes in real-time.
- **Disease Classification:** The system will analyze each mango image using a learning model to detect any signs of disease and classify the mango as healthy or diseased.
- **Disease Segmentation:** For diseased mangoes, the system segments the affected regions to determine the extent and area of infection accurately.
- **Question Answering and Export Insights:** The classified mango data will be sent to a Large Language Model (LLM), which, using Retrieval-Augmented Generation (RAG), will respond to the user queries and predict the most suitable export country and optimal marketplace.
- **Data Export and Reporting:** Users can generate and export detailed analysis reports produced by the LLM in various formats (e.g., PDF, DOCX) for documentation and future reference.

2.3. User Classes and Characteristics

Export edge supports diverse users in the export of mangoes, each differing in technical skill and frequency of use.

1. Mango Exporters

Description:

Use the system to get the classification of mangoes and their prices and the most suitable export country.

Characteristics:

- These users have little to medium technical knowledge but have an advanced understanding of mango export prices and the most suitable country.

- Use the platform frequently for the mango export process.
- Require access to analytical tools and explainability results for better decision-making.

2. System Administrators and Developers

Description:

This class includes project engineers, data scientists and technical personnel responsible for integrating new features or updating already implemented features.

Characteristics:

- High technical proficiency in software, AI, and edge devices.
- Responsible for system configuration, security, and data integrity.
- Use the system as needed for maintenance, upgrades, and model testing rather than for analysis and decision-making.

2.4. Operating Environment

Export edge operates on edge devices designed locally. It can only be accessed from the edge device it has been deployed on. The web interface for LLM question answering can be accessed through any modern web browser, independently of specialized local hardware, aside from a standard computing device and an active internet connection.

Software Environment

- **Operating Systems:** The computer vision and LLM will be running completely in Raspberry Pi OS or Ubuntu on Raspberry Pi. The web-based interface will be compatible with any OS, including Windows, macOS, and Linux.
- **Browsers:** The web-based interface will be compatible with Chrome, Firefox, Safari, and Edge (latest versions).
- **Model storage and versioning:** AI models are serialized using Joblib or ONNX for reproducibility and scalable deployment.
- **Data processing pipeline:** Employs pandas, numpy, and opencv for data cleaning, feature extraction, and image enhancement.

Hardware Environment

- Raspberry Pi will be used for the deployment of all the AI models, and they will be running locally without any other deployment.
- Any standard desktop, laptop with at least 8 GB RAM for accessing the web interface.

External Dependencies

- Market websites (e.g. AMIS) for market data, PyTorch and TensorFlow for deep learning, Hugging Face transformers for LLM finetuning

2.5. Design and Implementation Constraints

The development of the system faces the following constraints:

- **Market Data API Limitations:** Restricted data access depending on API rate limits and, availability of market price data.
- **Hardware Limitations:** The edge processing unit (e.g. Raspberry Pi or Jetson Nano) has limited computational resources, including CPU power, RAM and no or limited GPU acceleration, which may impact the performance of LLM.
- **Security Requirement:** The system must comply with data privacy guidelines, ensuring that mango images and generated reports are not shared with third parties. API keys used for LLM and database integration must be securely stored.

2.6. User Documentation

The following user documentation will be delivered with the project:

- **User manual:** A detailed guide on how to use the system, including question answering and report generation.
- **Online Help:** Context-sensitive help available within the application to assist users with specific tasks.
- **Video Tutorials:** Short, instructional videos demonstrating key product functionalities.

2.7. Assumptions and Dependencies

The following assumptions and dependencies are relevant to the project:

- **API dependency:** The availability and quality of market data rely on online APIs. Any interruptions in this service may impact the system functionality.
- **Third-Party Software:** Libraries such as PyTorch and OpenCV are assumed to remain actively supported and maintained throughout the software's lifecycle. Any major updates or deprecations could require adjustments in the system architecture.
- **Hardware Resources:** Requires sufficient GPU/CPU and memory resources for efficient model training.

3. External Interface Requirements

3.1. User Interfaces

The system will provide an intuitive and user-friendly **Graphical User Interface (GUI)** that allows users to easily interact with all main features of the application. The interface will follow a clean and consistent design standard to ensure smooth navigation and accessibility.

- **Screen Layout**

The screen layout for the vision module is as follows:

- A window showing the real-time video from the camera.
- A side window showing the classification result of mango images.

The screen layout for the web interface is as follows:

- A screen with a simplistic design and a light color theme responsive for desktops and tablets.
- A prompt bar for users to write the query.
- A button to upload the images of mangoes to be passed to LLM.
- **Standard Buttons and Functions**
 - Help: Access to help documentation.
- **Keyboard Shortcuts**
 - S: Save a frame from the real-time video.
 - Q: Quit the real-time video from the camera.
- **Error Messages**
 - Standardized error messages will be displayed in pop-ups if the image cannot be captured or if any prediction cannot be performed by AI models. Each error will guide resolution.

3.2. Hardware Interfaces

The software will interact with specific hardware components required for its operation. These interfaces define how the software communicates with the hardware, including input/output devices and processing units. The key hardware interactions are:

- **Raspberry Pi:** Used for running the classification and segmentation models.
- **Camera:** Used to capture images of mangoes on the conveyor belt.
- **Sorting Arm:** Used to sort mangoes based on the results of the classification model.
- **User's device:** The web-based application for LLMs will support a wide range of devices (laptops, desktops) with a focus on responsiveness and performance on both high-end and average systems. Additionally, the video of the conveyor belt will be displayed on the user's device, captured using the device's camera.
- **Input devices:** Keyboard and mouse interfaces for user interaction with the web application.

3.3. Software Interfaces

The system will integrate with various software components, APIs, and services:

- **Market Prices API**
 - Provides market-specific data for price prediction.
- **TensorFlow (Version 2.16+):**
 - Deep learning framework used to run the Mobilenet classification model.
 - Integration: Model inference pipeline for processing mango images and prediction.
- **Pytorch (version 2.8+):**
 - Deep learning framework used to run deeplab v3 segmentation model
 - Integration: Model inference pipeline for processing classified mango images and generating segmentation masks.
- **OpenCV (version 4.10+):**
 - Computer vision library used for processing images
- **Operating Systems**
 - The computer vision system will run on Raspberry Pi os for edge deployment.
 - The LLM-based web app will be compatible with Windows, macOS, and Linux-based operating systems.
- **Web Services**
 - APIs to allow external systems to interact with the LLM-based web app, including question answering, image uploading, and retrieving results.

- **Database**
 - **Database Management System:** The system shall use a simple database, like MySQL, PostgreSQL, for the storage of predictions, export results.
 - **Data Management:** The database should support JSON structures to efficiently store LLM responses.

3.4. Communication Interfaces

The system will rely on the following communication protocols:

- **HTTP/HTTPS:**
 - The primary communication protocol for interaction between the client-side application and the backend services.
 - HTTPS will be enforced for all communications to ensure secure data transfer.
- **RESTful API:**
 - The system will provide a RESTful API for external integrations with data sources for live market updates.
 - API documentation will be provided, including endpoint descriptions, required parameters, and response formats.
- **Security and Encryption:**
 - All data exchanges, especially LLM responses, will be encrypted using SSL/TLS.
- **Data Transfer Rules:**
 - Typical data transfer involves small JSON packets (<1 MB) containing image metadata and prediction results.
 - For image uploads to LLM, compression shall be applied to reduce bandwidth.
 - Since, user may upload multiple images, there will be optimizations for handling high data transfer rates. Batch processing of images will be employed to prevent timeouts during long-running operations.
- **Synchronization Mechanism:**
 - Synchronization between modules is sequential; the next module only starts when the previous one returns data, ensuring an accurate sequential process.

4. System Features

4.1 Mango Image Classification

4.1.1 Description and Priority

This feature enables the capture and classification of mango images using a deep learning model. It is a high-priority feature, as it is the base of the system. The benefit of providing accurate classification results is crucial for the next steps and for reducing post-harvest losses.

Priority Breakdown:

- **Benefit:** 9
- **Penalty:** 9
- **Cost:** 7
- **Risk:** 6

4.1.2 Stimulus/Response Sequences

1. **Stimulus:** The user starts capturing the video of the conveyor belt rolling mangoes.
 - **Response:** The system becomes active and starts scanning for mangoes.
2. **Stimulus:** Mango appears on the conveyor belt.
 - **Response:** The system takes a snapshot of the mango on the conveyor belt and starts processing it.
3. **Stimulus:** Processed mango goes to the classification model.
 - **Response:** The system runs the classification model to classify the mango.
4. **Stimulus:** Classification result is passed to the next step.
 - **Response:** The system saves the prediction and metadata.

4.1.3 Functional Requirements

- **REQ-1:** The system must allow the user to start recording the conveyor belt.
- **REQ-2:** The system must start scanning when the recording is started.
- **REQ-3:** The system must capture the frame when a mango appears on the conveyor belt.
- **REQ-4:** The system must preprocess the frame before passing it to segmentation models.
- **REQ-5:** The system must run the classification model on the processed frame.
- **REQ-6:** The system must save the classified mango image with metadata in the current directory.

4.2 Mango Image Segmentation

4.2.1 Description and Priority

This feature enables performing segmentation on classified mango images to create segmentation masks. This is a medium-priority feature.

Priority Breakdown:

- **Benefit:** 6
- **Penalty:** 5
- **Cost:** 7
- **Risk:** 3

4.2.2 Stimulus/Response Sequences

1. **Stimulus:** The classified mango image is saved with the metadata.
 - **Response:** The image will be passed to the segmentation model.
2. **Stimulus:** The system runs the segmentation model on the image.
 - **Response:** The system produces the segmentation masks and overlays them on the original image.
3. **Stimulus:** The system has produced the segmentation masks of the image.
 - **Response:** The system saves the segmentation mask with the metadata for further steps.

4.2.3 Functional Requirements

- **REQ-1:** The system must run the segmentation model on the classified image.

- **REQ-2:** The system must generate the segmentation mask overlayed on the original image.
- **REQ-3:** The system must save the segmentation masks for further processing and analysis.

4.3 LLM QA

4.3.1 Description and Priority

This feature allows users to have a chat with an LLM for the classified mangoes by uploading mango images. The user can ask generic questions about the guidelines for mango export or ask specific questions regarding classified mangoes. This feature is a Low priority.

Priority Breakdown:

- **Benefit:** 5
- **Penalty:** 5
- **Cost:** 8
- **Risk:** 5

4.3.2 Stimulus/Response Sequences

1. **Stimulus:** The system has generated the segmentation masks for the mango images.
 - **Response:** The images and metadata are stored in a directory.
2. **Stimulus:** A batch of images have been stored in a directory.
 - **Response:** The batch of images are uploaded to the LLM for price prediction.
3. **Stimulus:** The user asks a query regarding mangoes.
 - **Response:** The LLM generates response regarding the price and most suitable export country for mangoes.

4.3.3 Functional Requirements

- **REQ-1:** The system must save all the images of mangoes to be passed to LLM.
- **REQ-2:** The system must generate the response regarding prices and export country for mangoes.
- **REQ-3:** The system must generate a report for the mango processed.
- **REQ-4:** The system must let allow the user to download the generated report.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system must classify the mango after the image snapshot is captured within 1 second.
- The system must be able to generate the segmentation mask within 2 seconds.
- The price prediction module should have a **Mean Absolute Error (MAE)** of less than **10%** compared to the actual market price.

5.2 Safety Requirements

- The system must ensure data integrity that all image data, model files, and prediction results are **stored and processed securely**, preventing accidental loss or corruption.

- If the model fails to classify an image, it should **display an appropriate error message** rather than generating random or misleading outputs.
- The system must be able to revert to most recent valid state and notify user of the failure in case of malfunction.
- Guard rails must be implemented in the system so that the responses of LLM are safe for the users.

5.3 Security Requirements

- The system must save all image data, classification results, and price predictions **securely** to prevent unauthorized access or modification.
- The trained model files must be **read-only** during deployment to prevent accidental overwriting or tampering.
- Backup copies of models and datasets should be maintained to recover from data corruption or any malfunctions in the system.
- The system should comply with **institutional data security policies** and **AI research ethics** standards.

5.4 Software Quality Attributes

- **Availability:** The system must have an uptime of 99% , ensuring it can handle all the steps for continuous use during working hours.
- **Maintainability:** The codebase should be modular and well-commented to allow easy-updates, retraining or feature expansion.
- **Reliability:** The system should be able to work continuously for at least 7 hours without memory overflow.
- **Usability:** The user interface should be intuitive and easy to understand for non-technical users with a focus on simplicity and clarity in generating results.
- **Adaptability:** The system should support retraining with new mango datasets to adapt to additional varieties or quality parameters.
- **Testability:** The system must be easily testable, with unit tests covering at least 80% of codebase and automated testing workflows integrated into the development process.

5.5 Business Rules

- Data used for training must comply with **ethical and institutional data use guidelines** and should not contain copyrighted or personal content.
- The predicted price values are **indicative only** and should not be used for direct commercial transactions without human review.
- General users can only upload the mango images for prediction and view the price outputs.

6. Other Requirements

6.1 Database Requirements

- The system shall utilize a relational SQL database for consistent and structured storing of prediction results.
- **Schema Structure:** Each stored prediction result shall include metadata such as prediction time, prediction ID, user ID, and query logs.

- **Reliability:** The database must maintain transactional integrity (ACID compliance) and recover automatically from unexpected service interruptions.
- **Security:** Only authenticated and authorized roles may access database operations; direct database access is restricted to administrators.

6.2 Legal Requirements

- The system must comply with all relevant data privacy policies, including institutional data security policies for handling all the data.
- All pricing data must comply with licensing agreements from data providers such as the Agriculture Market Information Service, ensuring proper attribution where necessary.

6.3 Reuse Objectives

- The system should be designed with modularity in mind to allow reuse of components, such as the classification and segmentation modules, for similar agricultural products.
- Code reusability should be a priority with clear documentation and separation of concerns to allow the integration of the codebase into other projects or to introduce new features.

Appendix A: Glossary

Term	Description
AI (Artificial Intelligence)	The simulation of human intelligence in machines that can learn, reason, and make decisions.
API (Application Programming Interface)	A defined interface that allows different software systems to communicate and share data or functions.
CNN (Convolutional Neural Network)	A deep learning model used for analyzing visual data like images and videos.
MAE (Mean absolute Error)	A metric that measures the average difference between predicted and actual values in a model.
SSL/TLS: (Secure Sockets Layer/Transport Layer Security)	Protocols that secure data exchange over the internet through encryption.
QA (Questioning Answering)	An AI system that provides direct answers to user questions, often using natural language understanding.
Segmentation	A computer vision process that divides an image into parts or regions for easier analysis.

Classification	The task of assigning input data (like an input image or text) into one of several defined categories.
LLM (Large Language Models)	An advanced AI model trained on large text datasets that can understand and generate human-like language.

Appendix B: To Be Determined List

- TBD-1: Determine the batch size for batch processing of images and LLM upload.
- TBD-2: Determine if language localization support is needed or not.
- TBD-3: Determine if the LLM will be deployed locally or the cloud.